Software Engineeering

Lab -03

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Section:G

Problem 19: Personalized Meal & Diet Subscription Manager

1) **Selected Architectural Style : Microservices Architecture**

2) **Reasons for Selecting Microservices**
   i)**Separation of Personalized Meal-Planning Functions**: The system has distinct responsibilities that can be separated into independent services. For example, User Profile Service manages dietary information, while Meal Plan Generation Service uses this information to create personalized weekly meal plans. The Dietary Constraint Validation Service independently checks macronutrient requirements and allergen exclusions before the plan proceeds for dietitian review.
   ii) **Support for Performance Requirement:** The system requires a personalized weekly meal plan to be generated and displayed within 5 seconds for at least 95 out of 100 valid requests. With microservices, the Meal Plan Generation Service can be independently optimized and scaled without requiring the entire system to be scaled together

3) **Security Advantage**
   Microservices provide service-level isolation. Sensitive operations such as managing subscriber dietary profiles, subscriptions, and delivery information can be handled by separate services with controlled interfaces. Therefore, access to one service does not necessarily provide direct access to all other system functions.

4) **Performance Benefit**
   The architecture allows individual services to be independently optimized and scaled according to their workload.

Component Diagram:

